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Perspective

Linking activity dyshomeostasis and sleep disturbances in Alzheimer disease

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Abstract

The presymptomatic phase of Alzheimer disease (AD) starts with the deposition of amyloid-β in the cortex and begins a decade or more before the emergence of cognitive decline. The trajectory towards dementia and neurodegeneration is shaped by the pathological load and the resilience of neural circuits to the effects of this pathology. In this Perspective, I focus on recent advances that have uncovered the vulnerability of neural circuits at early stages of AD to hyperexcitability, particularly when the brain is in a low-arousal states (such as sleep and anaesthesia). Notably, this hyperexcitability manifests before overt symptoms such as sleep and memory deficits. Using the principles of control theory, I analyse the bidirectional relationship between homeostasis of neuronal activity and sleep and propose that impaired activity homeostasis during sleep leads to hyperexcitability and subsequent sleep disturbances, whereas sleep disturbances mitigate hyperexcitability via negative feedback. Understanding the interplay among activity homeostasis, neuronal excitability and sleep is crucial for elucidating the mechanisms of vulnerability to and resilience against AD pathology and for identifying new therapeutic avenues.

Sections

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Introduction

One of the first signs of symptomatic Alzheimer disease (AD) is the inability to remember recently learned information. On the basis of the degree of cognitive impairment, AD is often divided into three stages: the presymptomatic (also known as the preclinical) stage, characterized by normal cognitive ability; the prodromal stage, characterized by mild cognitive impairment (MCI); and the symptomatic, dementia stage, characterized by functional impairment[1] (Box 1). Ten to twenty years before the onset of cognitive impairments, amyloid-β (Aβ) begins to deposit in the cortex. This is considered the beginning of the presymptomatic stage[2] and is followed sequentially by hypometabolism, the accumulation of neurofibrillary tangles made up of hyperphosphorylated tau and hippocampal volume loss[3]. Although Aβ may cause cognitive dysfunctions independently from tau[4], dementia in individuals with AD is only noted when tau pathology has spread from the entorhinal cortex into the neocortex[5–8]. Subsequent progressive cognitive decline inevitably endangers most aspects of life and individuals with AD usually die within 5–12 years of diagnosis.

Around 30% of individuals who show neuropathological features of AD do not develop dementia[3]. Such resistance to AD pathology may arise as a result of a reduced pathological load (encompassing both initial Aβ deposition and subsequent tau aggregation) and/or an enhanced resilience of brain circuits, in which resilience is defined as the ability of a system to maintain homeostasis in the face of a perturbation. Over the past three decades, the predominant approach for the development of AD therapeutics has attempted to replicate the first of these possibilities by targeting amyloid pathology, a strategy extensively detailed in previous research[3]. Recent studies have shown that lecanemab, an anti-Aβ protofibril antibody, effectively reduces amyloid burden but results in only modest cognitive improvement during an 18-month phase III clinical trial[9]. More extended trials are thus necessary to determine the efficacy of amyloid pathology removal in treating dementia. However, in this Perspective, I focus on the other (relatively understudied) driver of resistance to AD: the modulation of resilience. In particular, I emphasize theoretical and experimental aspects of neural circuit vulnerability and resilience to early Aβ pathology.

Hyperexcitable patterns of neuronal network activity are detected in a fraction of individuals with AD[10,11]. These take the form of epileptiform activity, defined as paroxysmal sharp waveforms (spikes and sharp waves) on electroencephalogram (EEG), lasting 20–200 ms (ref. 10), that occur primarily during sleep[12,13]. This pathological activity is termed 'interictal epileptiform activity' when observed in individuals with seizures and 'subclinical epileptiform activity' when present in patients without a history of seizures. Similarly, in transgenic mouse models of familial AD (fAD), hyperexcitability emerges during low-arousal brain states, such as sleep and anaesthesia, and occurs before robust disruptions of cognition and global sleep patterns are present[14,15]. A connection between sleep disturbances and AD pathology has been extensively discussed. Notably, lower sleep efficiency predicts subsequently greater Aβ accumulation in individuals who are presymptomatic[16] and sleep deprivation increases extracellular levels of Aβ[17] and tau in humans and mice[18]. On the flip side, the presence of pathological forms of Aβ and tau can disrupt the sleep–wake cycle[19,20], potentially leading to a vicious cycle[21]. However, why hyperexcitability is limited to low-arousal states during the initial stages of AD and what the link is between this hyperexcitability and sleep disturbances remain open questions.

Answering these questions necessitates an updated theory of AD pathogenesis that considers changes to brain state-dependent network physiology. Five years ago, I, together with my colleagues, proposed the firing homeostasis and plasticity (FHP) hypothesis[22]. This hypothesis posits that the disruption of a balance between the homeostatic regulation of neuronal firing and Hebbian-like synaptic plasticity, which is integral to the cellular basis of learning and memory[23], is a driving force behind the manifestation of symptoms in the clinical stages of AD. Specifically, I suggested that dysregulation homeostatic sensors, set points or effectors can lead to dyshomeostasis of spike rate and pattern in AD. In this Perspective, this hypothesis is used to help us to understand brain-state-specific, AD-related resilience and vulnerabilities at the synaptic, neuronal and cognitive levels.

Box 1

The stages of Alzheimer disease pathogenesis

Alzheimer disease (AD) is the most common cause of dementia. Approximately 1–2% of AD is inherited in an autosomal dominant fashion and is characterized by early onset and a fast progression rate[201]. However, most AD occurs after the age of 65 and thus constitutes late-onset AD, a multifactorial disorder that is driven by numerous risk factors (including age, genetic mutations, vascular disease and infection)[202]. AD pathogenesis begins with a presymptomatic phase, during which abnormal amyloid-β biomarkers — but no tau pathology or cognitive impairment — can be detected[1]. Tau changes are subsequent to amyloid-β, and higher rates of tau accumulation occur during the progression of the disease to the prodromal phase (featuring mild cognitive impairment (MCI)) and, eventually, the symptomatic phase[203]. The 12-month progression rate from MCI to AD is 16.5% (ref. 204). Thus, most individuals with MCI remain stable, or even revert to normal cognition[2]. Those remaining convert to mild AD, characterized principally by functional changes such as a reduced ability to remember new information and a disrupted lifestyle. At the point of mild AD, progressive cognitive decline towards severe AD is irreversible.

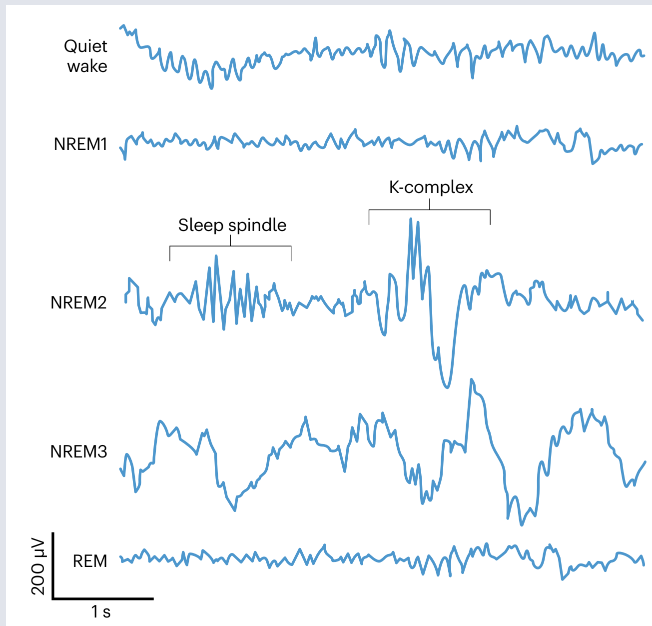
Neurodegeneration is region-specific and starts in mild AD in layer II of the entorhinal cortex. This is followed by neurodegeneration in the hippocampal CA1 region and neocortical association areas. Atrophy of the hippocampus and entorhinal cortex can predict the decline from MCI to AD[205,206] and exacerbates with continued cognitive decline[207,208]. Other brain regions, including the hippocampal dentate gyrus and the primary sensory cortices, are relatively resistant to neuronal degeneration until severe AD[209–211]. This is in contrast to normal ageing, in which the dentate gyrus is the most vulnerable region to degeneration[212].

Box 2

Sleep stages and rhythms

Active wakefulness is associated with low-amplitude, high-frequency cortical activity, whereas non-rapid eye movement (NREM) sleep is characterized by higher-amplitude, slow-wave activity patterns (schematic representations of typical activity patterns are shown in the figure). In humans, non-rapid eye movement 1 (NREM1) is the first stage of sleep and is characterized by a fairly asynchronous electroencephalogram similar to that seen quiet wake. NREM2, the second stage of sleep, is principally characterized by spontaneous occurrences of K-complexes and spindles. Next is NREM3, which is characterized by large amplitude, slow waves that spread throughout the brain, often originating from frontal regions. In mice, NREM sleep is typically not subdivided into stages. REM is a 'paradoxical' sleep stage that often emerges briefly after each bout of NREM and is characterized by some wake-like patterns of electrophysiological activity (such as a high-frequency, low-amplitude electroencephalogram) and other physiological signatures (such as its namesake eye movements). REM predominates later in sleep when sleep demand has already been partially quenched.

Several notable sleep rhythms are observed during these stages. Slow-wave oscillations reflect fluctuations of membrane potentials with a frequency lower than 1 Hz, with these membrane potentials toggling between depolarized UP states and hyperpolarized DOWN states[213]. Although mostly cortical in origin[214], thalamocortical projections are also likely to contribute to slow-wave activity in vivo[215]. Bidirectional transitions between UP states and DOWN states are generated by a transient, selective increase in synaptic excitation[216–218]. K-complexes are transient waves with a frequency between 0.5 Hz and 4 Hz and are associated with cortical DOWN states during NREM sleep[219,220]. There is a close connection between the formation of K-complexes and sensory stimulation[221]. Sleep spindles are isolated bursts of oscillatory neural activity with a frequency between 11 Hz and 16 Hz that occur during NREM sleep, reflecting interactions between the reticular thalamic nucleus and thalamocortical circuits[222]. Finally, sharp-wave ripples are composed of large amplitude, recurrent sharp-wave oscillations that are followed by synchronous high-frequency bursts of firing (~80–120 Hz in humans[196]), named ripples, that occur mainly during NREM sleep and quiet wake. They are best characterized in the hippocampus where they contain patterns of neuronal discharge that replay events in a compressed manner and contribute to learning[72,73].



In this Perspective, I initially explore pathophysiological phenotypes at the levels of brain states and local networks in individuals with AD and fAD mouse models. I posit activity dyshomeostasis in cortico-hippocampal circuits as a primary pathogenic mechanism underlying neuronal hyperexcitability during sleep in the presymptomatic disease stage and suggest that this hyperexcitability subsequently drives sleep and memory impairments in AD. Finally, I discuss the reciprocal relationship between sleep and activity homeostasis, proposing that sleep deficits may serve as an adaptive solution to mitigate hyperexcitability at presymptomatic AD stages. Separating the drivers of AD pathogenesis from the adaptive physiological responses that they trigger is crucial for developing disease-modifying approaches that enhance cognitive resilience.

Sleep phenotypes in AD

Sleep is defined as a reversible behavioural state in which the arousal threshold is increased, leading to a relative disengagement from the environment. Different vigilance states (which include wakefulness, non-rapid eye movement (NREM) sleep stages (NREM1, 2 and 3) and rapid eye movement (REM) sleep) are characterized by unique patterns of neuronal network activity and can be classified based on EEG and electromyogram recordings (Box 2). Sleep is homeostatically regulated and, accordingly, the pressure to sleep (sleep demand) is normally increased as time awake is extended[24–26]. Sleep quantity and quality show strong heritability[27,28], but are also impacted by experiences in the previous waking episode[24], such as learning[29].

It is well established that sleep characteristics undergo changes across the lifespan of a human. During healthy ageing, sleep becomes progressively more superficial, displaying less slow-wave activity, increased nocturnal awakenings, decreased nocturnal sleep time, increased frequency of daytime naps and an attenuation of the homeostatic response to sleep deprivation[30–32].

Acceleration or aggravation of such normal ageing processes may be part of AD pathogenesis, as has been extensively studied and reviewed[21,33]. Indeed, as in ageing, individuals with AD exhibit an impairment in the progressive enhancement of slow-wave activity that usually occurs during NREM sleep. Typical changes in sleep architecture in AD (when compared with age-matched control individuals) are as follows: the duration of NREM3 (refs. 34–43) and REM (refs. 37,40,41,44–46) sleep stages is significantly reduced; and NREM1 is conversely extended[40–42,47–49]. As in healthy ageing, individuals with AD also spend more time awake during the night and nap more during the day, leading to a more fragmented circadian rhythm[37,40,41,43,45,47,50,51]. Moreover, both increasing age and AD are associated with reductions across the frequency bands of slow waves, sleep spindles and K-complexes (Box 2) during NREM2 sleep[34,36,52–54]. Importantly, decreased NREM3 and REM sleep are associated with impairments in cognitive function in patients with AD[55]. Several sleep changes seem to be specific to AD and are not seen during normal ageing[33], such as slow-wave deficits that are specific to lower frequencies (<1 Hz) during NREM sleep in the medial prefrontal cortex[16,56] and a loss of EEG desynchrony during REM sleep[57,58].

Although these abnormal sleep phenotypes become more common and intensify after AD diagnosis, the extent to which sleep parameters are altered in individuals with MCI remains unclear. Some studies point to reduced duration of NREM3 sleep at this stage[39,54], whereas others suggest longer REM latency and a reduction in REM sleep duration[40,47,49,59–61]. In addition, individuals with amnestic MCI show fewer sleep spindles during NREM compared with control individuals[35,54]. Notably, robust architectural disruptions emerge following the conversion from MCI to AD.

Sleep disturbances are also observed in fAD mouse models, including an increased duration of wakefulness and reduced time spent in NREM sleep throughout the sleep–wake cycle, with more fragmented sleep[19,62,63]. These disruptions are believed to commence either at the onset or slightly before the beginning of cognitive decline and Aβ depositions in a frequently used amyloid precursor protein (APP)/presenilin-1 (PS1) fAD mouse model (which carry fAD-related mutations in the genes encoding APP and PS1)[19,62]. fAD models also exhibit a decrease in slow-wave activity during NREM sleep at an age corresponding to the onset of cognitive impairments[15,19].

Hyperexcitability during sleep and anaesthesia in AD

Given the changes in sleep duration and sleep-associated neuronal network activity in AD, it is important to understand how AD pathogenesis affects brain physiology across different vigilance states. A number of studies have revealed AD-related hyperexcitability that occurs specifically during low-arousal brain states, such as sleep and anaesthesia.

Epileptiform activity

Hyperexcitability of neuronal networks has been observed in individuals with MCI and AD, predominantly during sleep[10]. This usually takes the form of subclinical epileptiform activity, including spike-and-wave discharges. Such epileptiform activity has been detected in the cortex via scalp EEG or magnetoencephalography[12] during the night in ~42% of individuals with AD compared with ~10% of age-matched control individuals[12]. In another study, subclinical epileptiform discharges were detected in ~54% of individuals with AD without a history of epileptic seizures compared with ~25% healthy control individuals[64]. The true prevalence of subclinical epileptiform activity is, however, probably higher as this activity is not always picked up by scalp EEG or magnetoencephalography[13].

Notably, cortical epileptiform activity in individuals with AD occurs most often during NREM3, despite this stage being of shorter total duration than NREM2 and despite NREM3 being further reduced in AD[12,65]. Recordings made using intracranial electrodes suggest that epileptiform activity that is selective to sleep may occur more frequently in the hippocampus than in the neocortex in AD[13], but this requires further validation. Epileptiform activity has also been detected in an individual with MCI[13]; however, measuring epileptiform activity and its state dependency across a large population of susceptible subjects will be essential to understand its contribution to early stages of AD.

Epileptiform spikes or discharges have been detected in every fAD mouse model tested to date, including knock-in models that express physiological levels of APP with fAD mutations[14,15,66–70] (Fig. 1c). This sleep-selective hyperexcitability is an early response to the pathogenesis triggered by fAD mutations: epileptiform activity in the cortex and hippocampus begins in response to Aβ accumulation months before significant memory and sleep impairments[14,15] and as early as 5 weeks of age[14]. The firing rates of both excitatory and inhibitory neurons were transiently suppressed after each epileptiform event, reducing the probability of sharp-wave ripples[71], known to be important for memory consolidation[72,73]. In addition to sleep, anaesthesia also exposes epileptiform activity in fAD mice at an early age when sleep and memory are still normal[15,74]. As the disease progresses, epileptiform activity appears also during wake states[14,75] and this coincides with the stage at which sleep becomes compromised[19].

Abnormal firing rates and patterns

The effect of fAD mutations on neuronal firing can be readily estimated using in vivo electrophysiological single-unit recordings or calcium imaging techniques in transgenic mice or rats harbouring fAD mutations. In the hippocampal formation of wild-type mice in vivo, the mean firing rate (MFR) and the pattern of spikes of single neurons are important proxies for local network excitability and are regulated as a function of sleep homeostasis[76]. MFR normally decreases, whereas burstiness (propensity for transient, high-frequency trains of spikes) increases during low-arousal states of NREM[76] or anaesthesia[77] in comparison to wake states. Consistent with the fact that epileptiform discharges in early-stage AD are confined to low-arousal states, neuronal firing is also dysregulated in specific vigilance states in young fAD mice (Fig. 1).

During active wake, the mean and peak firing rates of neurons in hippocampal CA1, CA2/3 and dentate gyrus subfields[15,78–89] and in the entorhinal cortex[84,90–93] of freely moving fAD mice are normal from presymptomatic to late disease stages. Putative inhibitory interneurons in CA1 and the entorhinal cortex also have largely normal firing rates during wake[15,86,90–92,94], although a few studies have reported increased firing[89,90,92]. In many cases, firing rates during active wake are normal despite impairments to the tuning, remapping and/or stability of the spatially selective receptive fields of the neurons[78–85,87]. Thus, in hippocampus and entorhinal cortex, firing rates during wake express higher resilience to amyloid pathology than does the spatial code.

Under low-arousal brain states, such as anaesthesia, when spontaneous firing rates are normally very low, the firing rates of excitatory neurons in hippocampal CA1 (refs. 15,95–97) and in the entorhinal cortex[97] of fAD mice are higher than they are in controls. This is evident before memory deficits are observed. Studies using two-photon calcium imaging also demonstrate an increase in the fraction of hyperactive neurons under anaesthesia[98,99] that appears even before amyloid depositions[95] in fAD mice. The typical downregulation of firing rates that

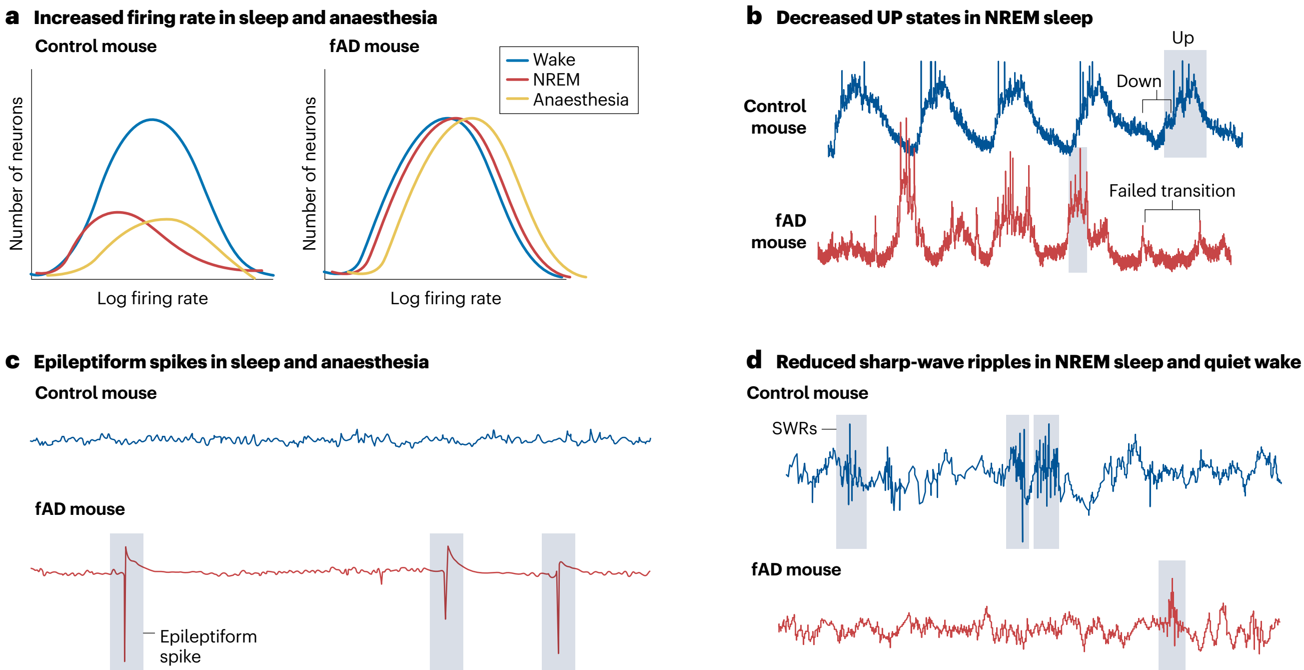


Fig. 1 | Network pathophysiology in mouse models of familial Alzheimer disease. a, The excitatory neuronal network in the hippocampal CA1 normally shows downregulation of firing rates in the transition from active wake to non-rapid eye movement (NREM) sleep or anaesthesia[15,74]. The figure shows a schematic representation of typical data showing that neurons in the amyloid precursor protein/presenilin-1 mouse model of familial Alzheimer disease (fAD) express similar firing rates in all vigilance states[15]. **b**, Slow-wave activity patterns comprise phase-locked transitions between depolarized UP states and hyperpolarized DOWN states in local populations of cortical neurons. In intracellular recordings from the somatosensory cortex of anaesthetized amyloid precursor protein/presenilin-1 mice (bottom trace), UP states are shorter than they are in control mice and transitions from DOWN to UP states often fail. **c**, Mutations associated with fAD increase the rate of epileptiform spikes[14,15,66–68,106,197–199]. **d**, fAD mice express reduced rate, amplitude and duration of sharp-wave ripples (SWRs) during quiet wake[86] and NREM sleep[71], compared with control mice. Part **b** is adapted with permission from ref. 200, Springer Nature Limited. Part **c** is adapted from ref. 15, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). Part **d** is adapted with permission from ref. 86, Elsevier.

occurs during the transition from active wakefulness to NREM sleep is also impaired in fAD mice during the presymptomatic disease phase[15].

High firing rates during sleep and anaesthesia may contribute to the appearance of epileptiform activity discussed in the previous section. Furthermore, this dysregulation of activity during sleep may result in a disruption of the diurnal regulation of Aβ[19] and could therefore be a risk factor for cognitive decline by increasing the production and/or release of Aβ[100–102] or other modulators of synaptic and intrinsic excitability (such as tau)[103,104]. Importantly, mechanisms that lower MFR have been found to suppress epileptiform discharges in intractable epilepsy[105] and fAD[15] mouse models in vivo. This suggests that an impairment to the mechanisms that regulate neuronal activity could underlie both the high firing rates and the epileptiform spikes seen in models of AD.

Hyperexcitability and memory decline

The rate of epileptiform spikes correlates with cognitive decline in individuals with AD[64] and those with subclinical epileptiform activity show accelerated AD progression[12,64]. Preclinical studies indicate that the atypical antiepileptic drug levetiracetam can reduce the rate of epileptiform spikes and improve memory in fAD mice[106]. Furthermore, a low dose of levetiracetam improved memory function in individuals with amnestic MCI[107]. In a recent randomized, double-blinded, placebo-controlled clinical trial, levetiracetam produced a performance improvement in spatial memory and executive function tests in a subpopulation of individuals with AD with comorbid epileptiform activity, although it did not affect the primary outcome[108]. Thus, strategies to target and suppress epileptiform activity in individuals with AD may benefit greatly from studies that aim to understand the mechanisms underlying subclinical epileptiform activity, specifically that occurring during low-arousal brain states.

The research in individuals with AD and animal models suggests a correlation between the rate of epileptiform discharges and cognitive function; however, a causal relationship between these parameters remains to be established. To address this question, a recent study focused on the neural circuit underlying hyperexcitability during low-arousal brain states in the presymptomatic stage in APP/PS1 mice. The thalamic nucleus reuniens was identified as a crucial brain region driving both firing rate dyshomeostasis and associated epileptiform activity in CA1 (ref. 74). Tonic deep brain stimulation (DBS) of the nucleus reuniens restored firing rate homeostasis and suppressed epileptiform activity during anaesthesia, preventing anaesthesia-induced impairments in short-term synaptic plasticity and working memory. Importantly, DBS of the nucleus reuniens during the presymptomatic disease stage also prevented age-dependent memory decline as the disease progressed.

Activity homeostasis, sleep and AD

The findings described in previous sections suggest that epileptiform discharges during low-arousal states drive the pathogenesis of AD. Here, I propose that this form of hyperexcitability stems from activity dyshomeostasis[15]. In the following sections, I will introduce a theoretical framework aimed at elucidating the breakdown of normal activity homeostasis within local neural circuits during sleep. Furthermore, I will explore the connection among activity dyshomeostasis, hyperexcitability and disruptions in sleep patterns.

Theoretical framework

To understand the sequence of events that takes place during the presymptomatic stage of AD and eventually progresses into cognitive deficiencies, it is useful to use the theoretical framework of homeostatic regulation and leverage the concepts of control theory[109] (Box 3). The concept of homeostasis, rooted in the seminal contributions of Claude Bernard, Walter Cannon and James Hardy, pertains to the mechanisms maintaining physiological variables within a dynamic range around a 'set point'[110–112]. Using principles from engineering control theory provides insight into the homeostatic regulation of neuronal activity, characterized by three essential features[109] (Box 3): a set point defining the output that the system must revert to after a perturbation; sensors detecting deviations from the set point; and homeostatic effectors employing negative feedback to precisely retarget regulated variables to the set point. The synaptic and intrinsic mechanisms driving these negative feedback loops are collectively termed homeostatic plasticity[113,114]. The FHP hypothesis proposed that AD risk factors impair important components of the homeostatic machinery that regulates neural activity[11,22], resulting in suboptimal homeostatic solutions that sacrifice some forms of memory.

There are several key assumptions that form the basis for the FHP. The first is that lowering homeostatic activity set points in central neural circuits can promote survival. This is supported by studies suggesting that longevity is promoted by transcriptional mechanisms that result in a reduction in activity and are conserved across species[115]. Second, it is assumed that multiple sensors coexist and allow activity set points to be maintained at different temporal scales[116,117]. In support of this, some homeostatic mechanisms have been shown to operate at the scale of

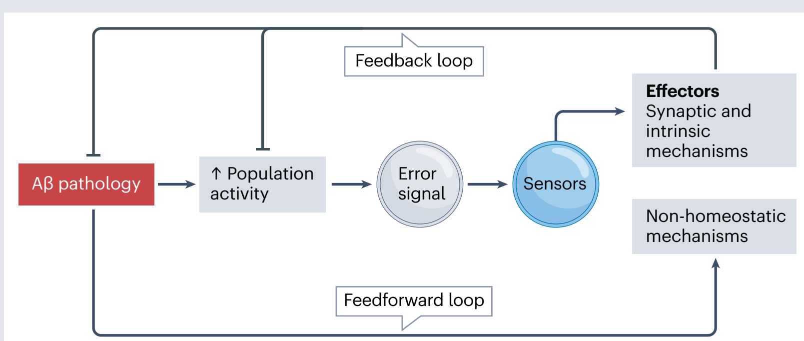
Box 3

Homeostatic regulation of activity in central neural circuits

In the framework of control theory, the average level of neuronal activity is typically preserved at a specific set point. Numerous studies in neural networks ex vivo[105,109,119,223,224] and in vivo[120,121,225] have demonstrated that the mean firing rate (MFR) of a neuron or neuronal population returns to baseline following a chronic perturbation that drives and increases or decreases in firing. Thus, MFR can be classified as a physiological variable that undergoes homeostatic set-point regulation. Homeostatic regulation of any physiological variable involves three main components: a set point that dictates the output of the network, sensors that detect deviations from this set point, and homeostatic effectors that return the regulated variable to a set-point value through negative feedback[109] (see the figure). I have previously proposed an integrative homeostatic network, composed of different homeostatic modules, that enables functional stability at different spatial scales, from single cell to cellular network, and protects from neurodegeneration[11]. The main core functional integrative homeostatic network modules are firing, genome, proteome, calcium, energy and immune system homeostasis[11].

Perturbations to brain activity can be categorized into two classes based on their effects on neural activity. Most perturbations change MFR only transiently because they do not impair the essential homeostatic modules (denoted as non-homeostatic mechanisms in the figure). Examples of such perturbations include a chronic increase in the extracellular GABA concentration induced by inhibition of GABA transporters[119] and a chronic increase in the extracellular concentration of glutamate induced by inhibition of glutamate transporters[105] in mouse hippocampal cultures. These perturbations inhibit and potentiate MFR, respectively, but result in MFR recovery typically after 2 days. Amyloid-β (Aβ) inhibits glutamate transporters[132], thus leading to an increase in glutamate concentration and hyperactivity (the feedforward loop in the figure). Homeostatic mechanisms, such as synaptic loss[175], mediate a feedback loop, aiming to renormalize Aβ-induced hyperactivity at the early disease stages. By contrast, some perturbations disturb essential components of the homeostatic machinery such that their effect on MFR is persistent and does not elicit a compensatory response. For example, TREM2 (ref. 175) and fAD[15,128] mutations impair homeostatic response to pathological Aβ forms.

Recent experimental work has identified several essential components of the MFR homeostatic machinery in central neural circuits. These include $GABA_B$ receptors, which are required for the adjustment of the MFR to compensate for hyperactivity[226]; insulin-like growth factor 1 receptor[149] and SH3 and multiple ankyrin repeat domains 3 (SHANK3)[122], which are required for the homeostatic responses to inactivity; and the mitochondrial enzyme dihydroorotate dehydrogenase, which is involved in determining the MFR set point[105].



minutes (such as the fast homeostatic process demonstrated at central synapses[118]), whereas others operate over the course of days[119–121]. Third, it is proposed that a failure to restore neuronal activity when it is perturbed can cause brain disorders. This is supported by recent studies showing that fAD mutations[15] and mutations in the autism spectrum disorder-related *Shank3* (ref. 122) and amyotrophic lateral sclerosis-related *Scnn1a*[123] genes also disrupt the homeostatic response to activity perturbations in mice. Fourth, it is assumed that there is normally a large number of degenerate solutions for stabilizing activity of neurons and neural circuits[124,125]. In AD, early pathogenic mechanisms may limit or alter this solution space, thus impacting the resilience of neural circuits to future perturbations[15,22]. Furthermore, a previous history of perturbations may be hidden in the activity pattern of a network under baseline conditions but revealed empirically by subsequent perturbations[117,126]. For example, anaesthesia exposes pathology in young fAD mice that is otherwise 'invisible'[15]. Finally, the FHP holds that synaptic plasticity can be sacrificed to maintain activity homeostasis. In support of this, neuronal networks have been shown to alter short-term synaptic plasticity to maintain the distribution of firing rates[119].

The FHP hypothesis can be used to gain insights into two basic questions: why low-arousal states (such as sleep and anaesthesia) are susceptible to hyperexcitability in AD and how activity dyshomeostasis, hyperexcitability and sleep disturbances may be interrelated.

Impaired activity homeostasis in AD

Recent findings support the idea that ageing and the early stages of AD are associated with impairments in homeostatic regulation. Neuronal activity, as estimated by in vivo somatic calcium imaging in V1, renormalizes after it is elevated in response to sensory stimulation in young mice, but remains elevated in aged mice[127]. This observed hyperactivity is associated with an increased synaptic excitation to synaptic inhibition ratio that results from a strengthening of the connections between excitatory neurons and disrupted cognitive function. In another study, longitudinal spike recordings in cultured hippocampal neurons revealed that fAD mutations disrupt homeostatic changes in the firing rate in response to chronic activity perturbations[15]. Specifically, the restoration of baseline MFR was disrupted during a perturbation that resulted in chronic hyperactivity (decrease in inhibition) but preserved during a perturbation that resulted in hypoactivity (increase in inhibition). Homeostatic plasticity at excitatory synapses has been shown to be impaired by fAD mutations[128] and pathological forms of Aβ[129,130], and this may contribute to the dysregulated firing rate homeostasis in AD models. These data support the idea that Aβ disrupts the homeostatic machinery and thus increases the vulnerability of neural circuits to perturbations[22]. Although a decrease in tau levels mitigates hyperexcitability in response to a perturbation in a fAD model[131], the extent to which pathological tau forms impede activity homeostasis has yet to be examined.

Mechanisms of impaired activity homeostasis during sleep

The dysregulation of activity that has been observed specifically during sleep or anaesthesia in response to amyloid pathology[14,15,71,95,98,132] may arise from impairments in various aspects of homeostatic regulation. Below, I discuss theoretical perspectives that focus on three specific aspects of the homeostatic framework that may contribute to this understanding.

Dysregulated error signal. In the homeostatic framework outlined in the FHP hypothesis, the 'error signal' is the magnitude of the deviation of activity from its set point. Aβ increases glutamate release probability[133] and suppresses glutamate reuptake[132], and Aβ load has been shown to induce an acute increase in excitatory drive, excitation–inhibition ratio, intracellular calcium levels and firing rate in mouse hippocampal and cortical neurons in vivo[132,134] and ex vivo[133,135]. Several mechanisms may explain why these changes result in a larger error signal during sleep than during wake and may thus surpass the homeostatic capacity of the system. For example, the specific neuromodulatory milieu that is present during sleep (including an increase in sleep-promoting and decrease in wake-promoting substances) could be hypothesized to enhance the effects of Aβ. Exploring this possibility is crucial, given that modulatory systems also undergo degeneration in AD[136]. Similarly, ectopically applied Aβ-containing extracts from the brain tissue of individuals with AD to mouse hippocampal neurons increase calcium events during anaesthesia through effects on receptors, including $GABA_A$, AMPA and NMDA receptors[132]. The conductances of those receptors are regulated by neuromodulators[137] and interstitial fluid ion composition[138] (both of which vary with sleep–wake states). Age-dependent AD risk factors also disrupt the diurnal fluctuation of Aβ levels[19,139], potentially impairing neural circuits during sleep. It is important to note that the sensitivity of firing rates to sleep–wake states, and thus the state-dependent effects of AD pathology, may vary anatomically[140].

Dysregulated homeostatic response. The homeostatic response is defined as the ability to compensate for a perturbation and is based on the availability and functionality of molecular sensors and effectors. Several theories relate to the role of sleep in regulation of homeostatic effectors. The synaptic homeostasis hypothesis posits that sleep has a crucial role in facilitating homeostatic plasticity mechanisms, thereby promoting synaptic renormalization in response to perturbations such as long-term synaptic potentiation during wake[141,142]. Given that pathological forms of Aβ have been associated with compromised long-term potentiation and augmented long-term depression[143], this theory suggests that compensatory responses taking place during sleep at the presymptomatic stage of AD might lead to an increase, rather than a decrease, in the synaptic strength or density. This would be predicted to heighten synaptic loss during wakefulness as a compensatory response to synaptic weakening associated with learning, an effect that would continue until the disease progresses sufficiently to cause disruption of the dependency of sleep duration and depth on duration of waking experience owing to the failure of sleep homeostasis[63,144]. This would potentially result in a weakened connection between wake and sleep states.

An alternative explanation suggests that sleep segregates homeostatic effectors that are induced in response to inactivity from those that are induced in response to hyperactivity, to maintain firing rates of neurons around a set point. Drawing from data obtained in the V1 area of the cortex during early development, it is suggested that sleep may activate homeostatic effectors in response to hyperactivity[145] but inhibit them in response to inactivity[146]. Consequently, these effectors are exclusively induced during active wakefulness[146]. Notably, fAD mutations have been shown to hinder homeostatic mechanisms that respond to hyperactivity but not those that respond to inactivity in cultured hippocampal neurons[15]. This observation suggests that disrupted sleep arising as a response to pathology could induce hyperexcitability by impairing homeostatic synaptic weakening. Additional investigations will be required to extending this perspective to the hippocampus and associated limbic structures in AD models in vivo.

At the mechanistic level, early Aβ pathology may affect several signalling pathways compromising the precision of compensatory responses taking place during sleep that normally restore network activity upon detecting a deviation from the set point[22]. For example, Aβ could disrupt homeostatic effectors such as the sleep-related neuromodulator, melanin-concentrating hormone, which is essential for MFR homeostasis[147]. Moreover, deficient insulin-like growth factor 1 (IGF1) receptor signalling has been observed in ageing and AD[148] and this may impede the ability of the system to restore network activity following a deviation from the set point as the IGF1 receptor is necessary for the upward MFR homeostatic response[149]. As IGF1 levels are lower in individuals with insomnia and in healthy individuals following sleep deprivation[150,151], it is possible that there may be deficient IGF1 release during sleep in AD, which may compromise homeostatic effectors.

Dysregulated homeostatic activity set point. Beyond merely facilitating homeostatic effectors, I propose that unique activity set points are established during sleep. In support of this idea, it is known that physiological variables, such as total excitatory synaptic strength and/or population spiking activity, are regulated around a homeostatic set point by sleep. Furthermore, it has been shown that, during NREM sleep, firing rate set points exhibit a narrower distribution[152–154], lower mean[15,76,140,154–156] and a critical regime of population activity[157] compared with active wakefulness. According to my hypothesis, during the presymptomatic phase of AD, activity set points remain preserved during wakefulness but undergo specific dysregulation during sleep[15]. In this scenario, the homeostatic mechanisms that take place during sleep are still functional, but operate with reference to a pathological set-point value. Set-point dysregulation could also be impacted by sleep-dependent effects of amyloid pathology. For example, AD-related pathogenic mechanisms may dysregulate sleep-dependent changes in the mitochondrial proteome[158] that have been shown to be essential for activity set point establishment[105]. Notably, cultures of neurons derived from APP/PS1 mice fail to reduce their activity set point in response to general anaesthetics, when compared with wild-type neurons[15], indicating deficient homeostatic mechanisms during low-arousal states.

An alternative perspective posits that NREM sleep homogenizes firing rate distributions through a combination of Hebbian-like and homeostatic processes[153]. According to this viewpoint, NREM sleep induces a homeostatic reduction in firing rates among a subpopulation of high-firing rate neurons, whereas causing an increase in firing rates among a subpopulation of low-firing rate neurons, owing to Hebbian plasticity. Given the compelling evidence on impairment of Hebbian-like long-term potentiation in AD models[143,159], it is anticipated to result in a decrease in firing among low-rate neurons, aligning with the theory proposed[153]. However, experimental data indicate an increased firing in this specific neuronal subpopulation[15], indicating a lack of correlation between long-term potentiation and modulation of low-firing neurons by NREM sleep in AD.

These theories converge on the idea that sleep has a role in the homeostatic regulation of neuronal activity, but diverge in their specific suggestions about what this regulation entails and how it occurs. Nonetheless, these views are not mutually exclusive. Perhaps, sleep-induced mechanisms govern various statistical parameters of both sub-threshold (synaptic) and supra-threshold (spiking) activities through various sensors, enabling regulation at different temporal scales. For example, sensors monitoring a full sleep–wake cycle (24 h) might generate an error signal that corresponds to the amount of time previously spent awake (in line with the synaptic homeostasis hypothesis theory). Simultaneously, another sensor might detect an error signal generated over the duration of a sleep bout (minutes to hours), thus preserving activity balance on shorter timescales. Analysing how AD disrupts the role of sleep in activity homeostasis, even before a substantial deterioration of sleep architecture, warrants exploration in future studies.

Taken together, these findings underscore the intricate relationship between early AD pathology and the homeostatic regulation of neural activity across different brain states. However, a comprehensive understanding of how sleep governs activity homeostasis in neural circuits is essential to pinpoint the specific homeostatic mechanisms that contribute to sleep-restricted destabilization of neuronal activity during early AD.

From dyshomeostasis to sleep disruption

The exact cause of sleep deterioration in AD is unknown. AD-related pathology in sleep-regulating brain areas could potentially trigger sleep dysfunction[21]. This includes the thalamus, hypothalamus, brainstem and basal forebrain, which together regulate sleep and slow-wave activities[160]. These areas are understudied in the context of AD, yet dysregulation of the thalamus has been linked to sleep disturbances in fAD mice[161]. It is crucial to recognize that dysregulated activity patterns in cortical and hippocampal areas may also contribute to sleep disturbances. Although the hippocampus is generally not considered to generate or sustain global brain states, it could be hypothesized that hippocampal hyperexcitability during low-arousal states may generate local sleep impairments and give rise to aberrant neuronal activity patterns at a larger spatial scale. Indeed, anaesthesia induces hyperexcitability in the circuitry among the thalamic nucleus reuniens, hippocampal CA1 region and the medial prefrontal cortex, which leads to working memory impairments in fAD mice[74]. Moreover, AD-related pathogenic mechanisms may exacerbate hyperexcitability in arousal-promoting neurons in the lateral hypothalamus, causing sleep fragmentation in aged mice[162]. Whether AD-related hyperexcitability originates in cortico-hippocampal circuits or in subcortical sleep-related brain regions is an important question for future investigation.

An alternative explanation for the changes in sleep architecture seen in AD could be a reduction in the homeostatic set point that drives sleep need. One conceivable scenario is that impaired sleep functions might contribute to a decrease in sleep need. Beyond affecting activity homeostasis, AD may also disrupt other established homeostatic functions of sleep, such as Aβ homeostasis (which depends on the steady state between Aβ production and elimination). Synaptic and spiking activities have crucial roles in Aβ production and release[101,102,163]. Sleep, characterized by reduced synaptic[164] and neuronal activity[15,76], is associated with lower Aβ levels in the interstitial fluid, whereas sleep deprivation leads to increased Aβ levels[17,165,166]. However, this regulation is significantly impaired by fAD mutations, ageing and amyloid aggregation[19,166]. Diurnal Aβ fluctuations disappear in 6-month-old APP/PS1 mice, preceding a reduction in NREM sleep time[19]. Sleep also participates in regulating Aβ elimination through the glymphatic system in wild-type mice[167], but glymphatic transport is suppressed in fAD[168] and aged[169] mice, potentially compromising sleep-dependent Aβ clearance mechanisms in AD.

Future studies are necessary to determine whether other sleep-dependent functions, beyond neuronal activity and Aβ homeostasis, deteriorate before notable disturbances in sleep architecture are detected. For example, parietal fast spindles, involved in memory consolidation[170], exhibit decreased density at the MCI stage, preceding

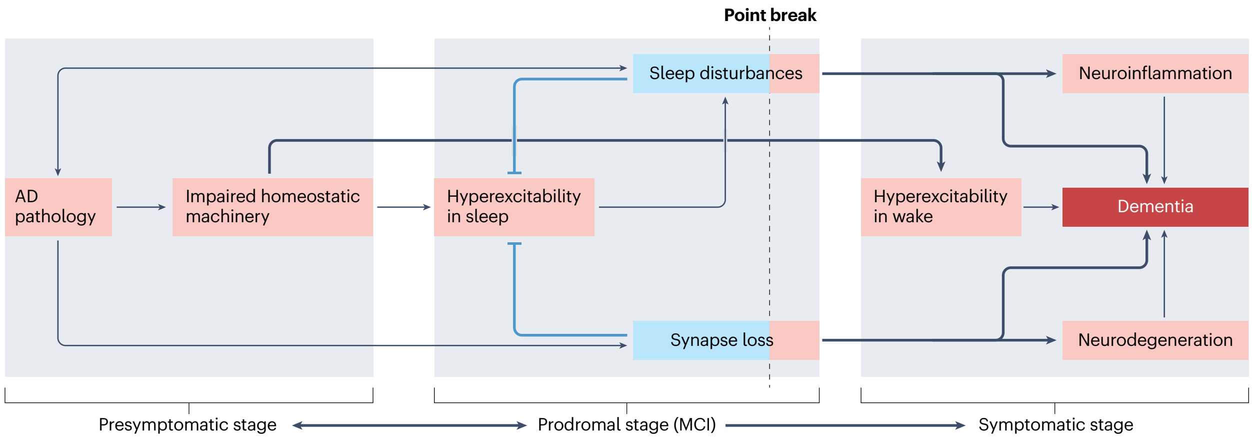


Fig. 2 | A new paradigm for the transition from presymptomatic to symptomatic phases of Alzheimer disease. According to the model put forward in this Perspective, accumulation of amyloid-β during the presymptomatic phase of Alzheimer disease (AD) impairs the homeostatic machinery, triggering the loss of activity homeostasis and hyperexcitability during low-arousal brain states, such as sleep and anaesthesia. Adaptive solutions that aim to minimize hyperexcitability and dyshomeostasis include microglia-mediated synapse loss and the suppression of sleep periods expressing epileptiform spikes. These mechanisms maintain overall activity during the 24-h sleep–wake cycle. Point break describes the point at which there is a loss of homeostatic regulation of population activity and at which these adaptive responses become detrimental. Additionally, point break is marked by a loss of activity homeostasis during wake, leading to memory impairments. The loss of homeostatic control in central neural circuits contributes to the failure of immune system homeostasis, leading to chronic neuroinflammation, exacerbating neurodegeneration and memory decline. MCI, mild cognitive impairment.

detectable changes in NREM sleep time[35]. Whether overall sleep need decreases during normal ageing[171] and the presymptomatic phase of AD remains to be determined.

Adaptive responses to activity dyshomeostasis

Synapse loss as a homeostatic response to hyperexcitability in early AD. Loss of synaptic contacts in both the neocortex and hippocampus is one of the major neuropathological hallmarks that correlate with cognitive decline in the brains of individuals with AD[172–174]. Although previous discussions have focused primarily on the negative impact of synapse loss on long-term synaptic plasticity and memory[159], it is also possible that synaptic loss may serve a homeostatic function by counteracting hyperexcitability in the early disease stages. Indeed, recent studies support a protective and compensatory role of synapse loss[133], demonstrating that the receptor TREM2 mediates accelerated synaptic elimination of hyperactive synapses by microglia and restores normal activity in response to Aβ-induced synaptic hyperactivity in cultured neurons. This effect is hindered by the R47H mutation in *TREM2* (ref. 175), which is known to increase the risk of AD twofold to fourfold[176,177]. This mutation increases synapse density in the cortex (probably as a result of reduced synaptic elimination by microglia[178]) and also augments epileptiform activity in fAD mice[179]. These results indicate that neuronal hyperactivity drives microglia to engulf synapses, which has a beneficial role in the homeostatic MFR regulation.

Weakening and loss of excitatory synapses may be a natural mechanism of homeostatic plasticity induced by sleep[141], reflected in structural and functional synaptic changes[180–183]. Sleep is also known to facilitate physiological microglia-mediated synaptic elimination[184], and the phagocytic functions of microglia may be either impaired[185] or enhanced[186] by sleep deprivation, depending on the brain area and the duration of sleep loss. Furthermore, microglia display highly motile processes that continually survey the brain parenchyma and transiently contact synaptic elements[187], a phenomenon known as microglia surveillance. Microglia surveillance is enhanced during anaesthesia compared with wakefulness[188,189]. As the contact of dendritic spines with microglia processes may facilitate synaptic elimination[187], enhanced surveillance may lead to augmented synaptic loss. However, the link among pathological forms of Aβ, microglia-mediated synaptic loss and brain states remains unexplored.

Future studies are required to investigate whether and how AD pathology and risk factors dysregulate microglia-mediated synaptic engulfment across sleep–wake states. For example, early amyloid pathology could accelerate physiological synaptic loss during sleep to restore activity homeostasis during the presymptomatic disease stages. At more advanced AD stages, increased amyloid load and microglia-specific AD risk factors may impair sleep and sleep-confined synaptic elimination. Moreover, AD-related pathogenic mechanisms may induce synaptic loss during wakefulness, resulting in memory disruptions. This hypothesis would suggest that therapeutic approaches aiming to prevent synapse loss could lead to epilepsy at early disease stages when synapse loss represents an adaptive strategy to maintain activity homeostasis.

Mitigating early hyperexcitability by sleep suppression. What is the functional significance of sleep changes during the presymptomatic phase of AD? There is a common perception that reduced or fragmented sleep is linked to poor disease prognosis, whereas increased sleep is considered protective. But is there solid evidence to support these assertions?

Sleep is an evolutionary mechanism optimized to the needs of a species. Importantly, species can adapt their sleeping patterns according to their current constraints and circumstances. For example, some mammals and birds transiently suppress sleep in response to external threats such as increased predation[190], whereas others (including humans) modify sleep patterns in response to internal threats such as infections, as part of a sickness behaviour that is considered an adaptive survival mechanism[191]. This indicates that sleep disturbances could be seen as both a beneficial response and a pathological effect. Thus, a radical idea is that impaired sleep may represent an adaptive response to AD pathology aimed at mitigating the cellular and molecular damage associated with hyperexcitability. This adaptive response may be triggered by the failure of local homeostatic mechanisms that are typically activated during sleep. Given that a high mortality rate is associated with epilepsy and seizures[192], the reduction in total sleep duration and fragmentation of sleep bouts seen in AD might be beneficial for survival by suppressing the daily rate of epileptiform spikes and preventing seizures. Thus, sleep deficits may be adaptive during the presymptomatic disease phase, representing the price that must be paid to abrogate a vicious cycle between Aβ and hyperexcitability.

Point break: transition to symptomatic AD

If sleep disturbances have evolved as an adaptive strategy to mitigate the vulnerability of sleep states to hyperexcitability during the presymptomatic disease stage, they become maladaptive during the transition to the clinical disease phase. An escalation in pathological burden may eventually surpass the capacity of the system for homeostasis, resulting in homeostatic loss not only during sleep but also during wake states[14,71]. I here use the term 'point break' to denote a pathophysiological threshold beyond which adaptive responses such as sleep disturbances and synaptic loss transform into maladaptive processes. This transformation leads to impaired memory consolidation[193] and a dysregulated immune response, evident in the shift towards increased inflammatory activity and decreased antiviral responses[194]. Another point break may occur when the homeostatic machinery can no longer prevent hyperexcitability during high-arousal wake states. Reaching this point break correlates with the loss of cognitive resilience and the transition from the prodromal to the symptomatic stages of AD (Fig. 2). Subsequent impairments of cognition and neurodegeneration would then be accelerated by the impaired systemic immune response and brain–immune imbalance[195], fostering chronic inflammation and potential irreversibility. Although we are beginning to understand the reciprocal relationships between sleep and activity homeostasis in local circuits in AD pathogenesis, much work is still required to fully elucidate its underlying principles and mechanisms.

Restoring activity homeostasis: a new promise

A significant challenge in AD research lies in distinguishing between the symptoms of AD and its underlying drivers. For example, if sleep disturbances serve as a compensatory mechanism for hyperexcitability, direct intervention in sleep homeostasis might have adverse effects, especially in the early stages of AD. The use of hypnotics, often referred to as 'sleep pills', to extend sleep in patients with AD may pose risks if sleep-restricted hyperexcitability is not concurrently reduced, potentially accelerating cognitive decline[12]. Similarly, instead of solely targeting memory deficits for symptomatic relief, identifying the elements of homeostatic regulation that are responsible for Aβ-induced hyperexcitability could offer a strategy to delay AD progression from the presymptomatic to symptomatic stages. Potential approaches grounded in the FHP theory may involve pharmacological and genetic therapies, such as reducing pathological activity set points through mitochondrial dihydroorotate dehydrogenase inhibition[15,105], rescuing homeostatic sensors via mitochondrial calcium uniporter activation[149] or activating homeostatic effectors through the signalling pathway induced by the sleep-promoting melanin-concentrating hormone[147]. Furthermore, exploring brain stimulation therapies opens avenues to suppress hyperexcitability in specific brain areas and brain states. Whether DBS in the prefrontal cortex during sleep[196] or in the thalamic nucleus reuniens[74] can prevent memory deterioration over an extended timescale by restoring activity homeostasis remains to be tested in patients with AD.

In conclusion, research centred on activity homeostasis in AD not only offers potential avenues for keeping the disease in a dormant state but also enhances our understanding of the interplay among synaptic plasticity, sleep and memory.

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